ABSTRACT

The Smart City Complaint Management System is a web-based application developed to digitalize and simplify the process of registering, monitoring, and resolving public complaints in an urban environment. The system provides a common platform where citizens can report civic issues such as water supply problems, electricity faults, sanitation issues, road damage, and other public grievances without visiting municipal offices physically.

The project is designed with multiple user roles, including Citizen, Department Admin, Field Worker, and Super Admin. Citizens can register, log in, lodge complaints, choose departments, and track the real-time status of their complaints. Department Admins can monitor complaints belonging to their department, assign tasks to field workers, and supervise resolution progress. Field Workers can view assigned tasks, update work status, and add completion notes. Super Admins have full control over the system, including managing users, departments, complaints, analytics, and audit logs.

A major strength of this project is its workflow-based complaint lifecycle management. Each complaint moves through stages such as Pending, In Progress, Resolved, and Closed, allowing transparent tracking from submission to completion. The system also stores status logs to maintain an audit trail of every complaint update. In addition, it includes a notification mechanism to inform users about complaint progress, along with support for email and SMS alerts, improving communication between citizens and authorities.

The database design of the system includes core entities such as roles, departments, users, complaints, task assignments, status logs, and notifications, which together support secure and organized data management. The project also provides reporting and analytics features, helping administrators understand complaint distribution by department and status. Overall, this system acts as an effective digital bridge between the public and city administration, improving service transparency, accountability, response speed, and operational efficiency in complaint

CHAPTER 1

INTRODUCTION

In today's rapidly expanding urban areas, effective management of public services and infrastructure has become increasingly important. With the rise in population, cities face numerous challenges such as water supply disruptions, electricity failures, improper waste management, and road damages. Traditional complaint handling methods, which involve visiting municipal offices or making phone calls, are often time-consuming, inefficient, and lack proper tracking mechanisms. This results in delays in resolving issues, poor communication, and reduced citizen satisfaction.

To address these challenges, the Smart City Complaint Management System is developed as a digital platform that enables citizens to register and track complaints easily through a web-based interface. The system eliminates the need for manual processes and provides a centralized platform where all complaints are recorded, monitored, and managed efficiently.

The application is built using PHP for server-side scripting, MySQL for database management, and HTML, CSS, JavaScript, and Bootstrap for designing an interactive and user-friendly interface. The system ensures secure data handling and structured storage of complaint information, user details, and status updates.

One of the key features of this system is its role-based access control, which allows different users to interact with the system according to their responsibilities. Citizens can submit complaints and check their status, Department Admins can review and assign tasks, Field Workers can update the progress after resolving issues, and the Super Admin manages the overall system operations.

1.2 FEASIBILITY STUDY

Feasibility study is an important phase in system development used to evaluate whether the proposed system is practical, efficient, and beneficial for implementation. The Smart City Complaint Management System, a web-based application designed for registering, tracking, and resolving public complaints, has been analysed based on the following feasibility aspects.

1.2.1 Technical Feasibility

The proposed system is technically feasible as it is developed using standard and widely used web technologies such as PHP, MySQL, HTML, CSS, JavaScript, and Bootstrap. These technologies support the development of a responsive and userfriendly web application with efficient data management. The system is designed with a structured database and role-based access control, allowing different users such as Citizens, Department Admins, Field Workers, and Super Admin to interact with the system according to their roles. The implementation of a workflow-based lifecycle (Pending, In Progress, Resolved, Closed) is technically achievable and ensures smooth complaint tracking.

Additionally, features like notifications, email/SMS alerts, and reporting tools can be integrated easily using available libraries and APIs, making the system technically reliable and scalable.

1.2.2 Economic Feasibility

The Smart City Complaint Management System is economically feasible as it is built using **open-source technologies**, which eliminates licensing costs. The development cost is relatively low compared to traditional systems. By digitizing the complaint process, the system reduces paperwork, manual effort, and administrative costs. It improves efficiency in handling complaints, reduces delays, and optimizes resource usage. The inclusion of automated features such as complaint tracking, notifications, and reporting helps in minimizing operational expenses in the long term.

1.2.3 Operational Feasibility:

system is operationally feasible as it provides a simple and intuitive interface for all users. Citizens can easily register complaints and track their status, while administrators can manage and assign tasks efficiently. The role-based system ensures proper coordination between users, and the workflow lifecycle helps in systematic complaint handling. The presence of status logs improves transparency, and the system enhances communication between citizens and authorities. Features such as real-time notifications and alerts ensure that users are updated about complaint progress, making the system more effective and user friendly in real-world operations

1.2.4 Legal Feasibility:

The system complies with legal and regulatory requirements related to data handling and privacy. It ensures that user information and complaint details are stored securely in the database. Proper authentication and authorization mechanisms are implemented through **role based access**, preventing unauthorized access. The system uses collected data strictly for complaint management purposes, ensuring ethical and legal usage.

CHAPTER 2

SYSTEM ANALYSIS

System analysis is an essential phase in the development of the Smart City Complaint Management System, as it involves a comprehensive study of the current complaint handling process and the identification of its limitations. In rapidly growing urban environments, the number of public issues related to water supply, electricity, sanitation, and road infrastructure is increasing significantly. However, the traditional methods used for managing these complaints are not efficient enough to handle the growing demand. This creates a gap between citizens and municipal authorities, resulting in delayed services and dissatisfaction among the public.The analysis of the current system reveals that there is a lack of proper coordination, transparency, and accountability in handling complaints. Citizens often do not receive timely updates about their complaints, and authorities face challenges in managing and tracking large volumes of data. There is no centralized platform to store and monitor complaint details, which leads to mismanagement and inefficiency.

2.1 AIM

The main aim of the Smart City Complaint Management System is to study and understand the existing manual complaint handling process used by municipal authorities, identify its limitations, and develop an efficient web-based application that simplifies the process of registering, tracking, and resolving public complaints.

2.1.1 Through system analysis, the goal is to:

Examine the current manual methods used for registering and managing public complaints and identify the challenges faced by citizens and municipal authorities in terms of delay, lack of transparency, and poor communication. Define the functional and nonfunctional requirements of the proposed system and design a structured web-based platform that enables citizens to submit complaints easily and track their status in realtime.

Establish a role-based system that includes Citizen, Department Admin, Field Worker, and Super Admin to ensure proper coordination and task management. Implement a workflowbased lifecycle such as Pending, In Progress, Resolved, and Closed to ensure systematic handling of complaints. Integrate features like notifications, email/SMS alerts, and reporting tools to enhance communication, improve efficiency, and support better decision-making in urban complaint management.

2.2 EXISTING SYSTEM

In the existing traditional system, public complaints related to urban services such as water supply, electricity, sanitation, and road maintenance are mainly handled through manual processes. Citizens are required to visit municipal offices, make phone calls, or submit written applications to register their complaints. This approach has been followed for many years, but it is not suitable for the modern digital environment where faster and more efficient solutions are required.

The manual complaint handling system lacks proper organization and coordination between different departments. Complaints are often recorded in registers or basic systems without a centralized database, making it difficult to store, retrieve, and manage information effectively. Citizens do not receive proper updates regarding the status of their complaints, which reduces transparency and trust in public services. As the number of complaints increases, managing them manually becomes more complex and inefficient, leading to delays and miscommunication.

2.2.1 Limitation of the Existing

Limited Accessibility Citizens must physically visit municipal offices or rely on phone calls to register complaints, which is inconvenient and time-consuming, especially for people living in remote areas. Lack of Transparency There is no proper system for tracking the status of complaints, and citizens are often unaware of the progress or resolution of their issues. Manual Record Keeping Complaint details are maintained manually, increasing the chances of errors, duplication, and data loss.

Delayed Response Complaints are not processed in a structured manner, leading to delays in assigning tasks and resolving issues. Poor Communication There is a lack of coordination between departments and field workers, resulting in miscommunication and inefficiency. No Centralized System The absence of a centralized database makes it difficult to manage, monitor, and analyze complaint data effectively.Lack of Reporting and Analysis The existing system does not provide any tools for generating reports or analyzing complaint trends, making it difficult for authorities to make informed decisions. Inefficient Workflow There is no defined workflow or lifecycle for handling complaints, which results in unorganized processing and reduced accountability.

2.3 PROPOSED SYSTEM

The proposed Smart City Complaint Management System is a web-based application designed to provide an efficient, transparent, and user-friendly solution for managing public complaints. Unlike the existing manual system, this application allows citizens to register complaints online from anywhere, eliminating the need to visit municipal offices physically. The system provides a centralized platform where all complaint data is stored in a structured MySQL database, ensuring proper organization and easy access to information.The system is developed using modern web technologies such as PHP, MySQL, HTML, CSS, JavaScript, and Bootstrap, which ensures a responsive and interactive user interface. It incorporates rolebased access control, allowing different users such as Citizens, Department Admins, Field Workers, and Super Admin to perform their respective tasks efficiently.

The system follows a workflow-based lifecycle, where complaints move through stages such as Pending, In Progress, Resolved, and Closed, ensuring systematic handling and tracking. Additionally, the system includes features such as real-time notifications, email and SMS alerts, and reporting tools that help in improving communication and decision-making.

2.2.1 Features of the Proposed System

1 .User-Friendly Interface:

The system is designed with a simple and responsive interface using Bootstrap, making it easy for citizens and administrators to access and use the application without requiring technical knowledge.

2. Complaint Registration:

Citizens can easily submit complaints online by providing details such as category, description, and location, which are stored securely in the database.

3.DataSecurity:

All user information and complaint data are stored securely with proper authentication and authorization mechanisms to prevent unauthorized access.

4.Real-TimeStatusTracking:

Users can track the progress of their complaints in real-time through different stages such as Pending, In Progress, Resolved, and Closed.

5.Multi-UserAccess:

The system supports multiple user roles including Citizen, Department Admin, Field Worker, and Super Admin with appropriate access permissions.

6.Complaint Assignment:

Department Admins can assign complaints to field workers based on category and location, ensuring efficient handling of issues.

7.Notifications and Alerts:

The system provides real-time notifications along with email and SMS alerts to keep users informed about complaint updates and status changes.

8.Reporting and Analytics:

Administrators can generate reports based on complaint categories, status, and time period, which helps in analyzing trends and improving decision-making

2.4 MODULES

The Smart City Complaint Management System is divided into several interconnected modules that work together to provide a complete and efficient solution for managing public complaints. Each module is designed to handle specific functionalities such as user management, complaint handling, task assignment, status tracking, and reporting. These modules ensure smooth system operation, better coordination between users, improved transparency, and reliable system performance.

2.4.1. User Registration and Login Module

This module allows new users to register and existing users to log into the system securely. It supports different types of users such as Citizens, Department Admins, Field Workers, and Super Admin.

The system provides essential functions such as new user registration using basic details, secure login with proper authentication, password management, and profile maintenance for all user roles. These features ensure that users can safely access and manage their accounts within the system. The main purpose of these functionalities is to maintain a secure record of users and provide controlled access based on role-based permissions. This helps in protecting sensitive information and ensures that only authorized users can perform specific actions within the system.

2.4.2 Complaint Management Module

This module is responsible for handling the submission and storage of complaints raised by citizens. It allows users to enter details such as complaint category, description, and location.

The system includes important functions such as complaint registration, automatic generation of a unique complaint ID, storing complaint details securely in the database, and allowing users to view the status of their submitted complaints. These features ensure that all complaints are properly recorded and can be easily tracked at any time. The main purpose of this module is to provide a centralized system for recording and managing all complaints efficiently. It helps in organizing complaint data, reducing manual work, and ensuring quick access and better handling of user issues.

2.4.4 Complaint Assignment Module

The system provides key functions such as viewing newly registered complaints, assigning them to appropriate field workers, and managing task allocation efficiently. These features help administrators monitor incoming issues and distribute responsibilities based on availability and expertise. By organizing and assigning tasks systematically, the system reduces delays and avoids confusion in handling complaints. The main purpose of this module is to ensure that each complaint is handled by the appropriate personnel, leading to faster resolution and improved service efficiency.

2.4.5 Status Tracking Module

This module manages the lifecycle of complaints by updating their status at different stages. The system includes important functions such as updating the complaint status at different stages like Pending, In Progress, Resolved, and Closed. It also maintains detailed status logs with timestamps to keep a complete record of all updates. Additionally, users are allowed to track the progress of their complaints in real time. These features help in monitoring the lifecycle of each complaint effectively. The main purpose of this module is to ensure transparency and proper tracking of complaint progress, enabling both users and authorities to stay informed about the status and actions taken.

2.4.6 Notification Module

This module provides communication between the system and users by sending updates about complaint status. The system provides essential functions such as sending real-time notifications, email alerts, and SMS messages to inform users about complaint updates and task assignments. These communication features ensure that users receive timely information regarding the status of their complaints without needing to check the system frequently. Notifications are also sent to authorities and field workers to keep them updated on assigned tasks. The main purpose of this module is to keep users wellinformed and to improve communication between citizens and authorities. This enhances transparency, reduces response time, and ensures a more efficient complaint management process.

2.4.7 Reporting and Analytics Module

This module helps administrators analyze complaint data and system performance. The system includes functions such as generating reports based on complaint category, status, and date, as well as analyzing trends and overall performance. These reports provide a structured view of complaint data, helping authorities understand the number and type of issues reported over a period of time. By examining patterns and trends, the system enables identification of frequently occurring problems and areas that require immediate attention. The main purpose of this module is to support effective decision-making and improve service efficiency by providing accurate and meaningful insights.

2.4.8 Admin Control Module

This module allows the Super Admin to manage the overall system and monitor all user activities effectively. It includes important functions such as managing users, monitoring system operations, controlling system settings, and accessing complete reports. The Super Admin has full authority to oversee and regulate the entire system, ensuring that all processes run smoothly. By having access to detailed reports and system controls, the admin can identify issues, track performance, and make necessary improvements. The main purpose of this module is to ensure smooth operation, maintain system integrity, and provide full control over all system activities.

CHAPTER 3

SYSTEM SPECIFICATION

The system specification defines the technical requirements and environment needed for the development and operation of the Smart City Complaint Management System. This web-based application is designed to function efficiently using standard hardware and software components, ensuring smooth performance, reliability, and scalability. The system is developed using modern web technologies such as PHP for server-side scripting, MySQL for database management, and HTML, CSS, JavaScript, and Bootstrap for designing a responsive and user-friendly interface.

The application follows a client-server architecture, where users interact with the system through a web browser, and all processing and data management are handled on the server side. The system supports multiple user roles such as Citizen, Department Admin, Field Worker, and Super Admin, each having specific functionalities controlled through rolebased access. It also incorporates a workflowbased complaint lifecycle including Pending, In Progress, Resolved, and Closed stages, ensuring proper tracking and management of complaints. The system is designed to handle features such as complaint registration, status tracking, assignment, notifications, and reporting efficiently. It ensures secure data storage, fast processing, and reliable communication between users and the system.

3.1 HARDWARE REQUIREMENT

The hardware requirements for the Smart City Complaint Management System are minimal, as it is a web-based application that can run on standard computer systems.A computer or laptop with a minimum of Intel Core i3 processor or equivalent is sufficient for development and usage. The system should have at least 4GB RAM and 500GB hard disk for storing application files and database. A stable internet connection is required for accessing the system. Input devices such as a keyboard and mouse are necessary, and a monitor is required for display purposes.

3.1.1. Server-Side Hardware Requirements

The server hosts the backend application, manages data processing, and stores all records such as user information, product details, and transaction logs. Hence, it requires a reliable and highperformance setup.

3.1.2 Hardware Requirements

- **Processor:** Intel Core i3 / i5 or higher
- **RAM**: Minimum 4 GB (Recommended 8 GB Storage: 250 GB HDD or 128 GB SSD and

  above
- **Display**: 14" or 15.6" HD / Full HD screen
- **Network**: Wi-Fi enabled with stable internet
- **Battery Backup**: Minimum 4 hours
- **Optional Devices**: External mouse, printer, UPS

3.2 SOFTWARE REQUIREMENTS

- **OS:** Windows 10/11, Linux, macOS (Client), Windows Server/Linux Server (Server)
- **Web Browser:** Chrome / Firefox / Edge
- **Frontend:** HTML5, CSS3, JavaScript, Bootstrap/Tailwind CSS
- **Backend:** PHP, Apache/Nginx, MySQL/MariaDB, phpMyAdmin

- **Development Tools:** VS Code / PhpStorm, Git, Photoshop/Canva

3.3 SOFTWARE DESCRIPTION

The software requirements include both development and execution environments needed for building and running the system. The application is developed using PHP as the backend programming language and MySQL as the database management system. Frontend technologies include HTML, CSS, JavaScript, and Bootstrap for creating a responsive and user-friendly interface. The system can be developed using tools such as XAMPP or WAMP server for local hosting. The application can run on any modern web browser such as Google Chrome, Mozilla Firefox, or Microsoft Edge. The operating system can be Windows, Linux, or macOS. Additional tools such as Visual Studio Code or Notepad++ can be used for coding and development.

3.3.1. System Overview

The Smart City Complaint Management System is a web-based application designed to provide an efficient platform for registering, tracking, and resolving public complaints related to urban services such as water supply, electricity, sanitation, and road maintenance. The system enables citizens to submit complaints online and monitor their status, while municipal authorities can manage and resolve issues effectively through a centralized system. The application uses modern web technologies and follows a role-based access approach, allowing different users such as Citizens, Department Admins, Field Workers, and Super Admin to interact with the system efficiently. It ensures proper coordination and transparency through a structured workflow.

3.3.2 System Users

The system is designed for four main types of users.

1.Citizens:

Citizens can register, log in, submit complaints, and track the status of their issues. They also receive updates and notifications regarding complaint progress.

2.DepartmentAdmin:

The admin manages complaints by reviewing, verifying, and assigning them to field workers. They also monitor complaint progress and ensure timely resolution.

3.FieldWorker:

Field workers are responsible for resolving assigned complaints and updating their status in the system.

4.SuperAdmin:

The super admin controls the entire system, manages users, monitors activities, and generates reports for analysis.

CHAPTER 4

SYSTEM DESIGN

The System Design of the Smart City Complaint Management System provides a clear blueprint for implementing the entire project. It defines how different components of the system interact with each other and ensures smooth and efficient functioning. The design focuses on creating a user-friendly, secure, and scalable platform that allows citizens to register complaints and track their status easily. design emphasizes efficiency by ensuring fast response time and proper handling of complaints without delays. The system is built in such a way that multiple users can access it simultaneously without affecting performance. It also ensures proper data flow between modules such as complaint registration, assignment, and status tracking.

Security is a major consideration in the system design. All user data and complaint details are stored securely, and access is controlled through role-based authentication. This ensures that only authorized users such as admins and field workers can manage complaints. Scalability is another important aspect of the system. The design supports future enhancements such as adding more departments, integrating mobile applications, or expanding to larger cities. This makes the system flexible and adaptable to growing requirements.

The system also focuses on providing a better user experience. Citizens can easily submit complaints, receive updates, and track progress in real time. The interface is designed to be simple and accessible for users of all age groups. By following this design, the system ensures improved communication between citizens and municipal authorities, increased transparency in complaint handling, and better accountability in resolving public issues.

4.1 ENTITY RELATIONSHIP DIAGRAM

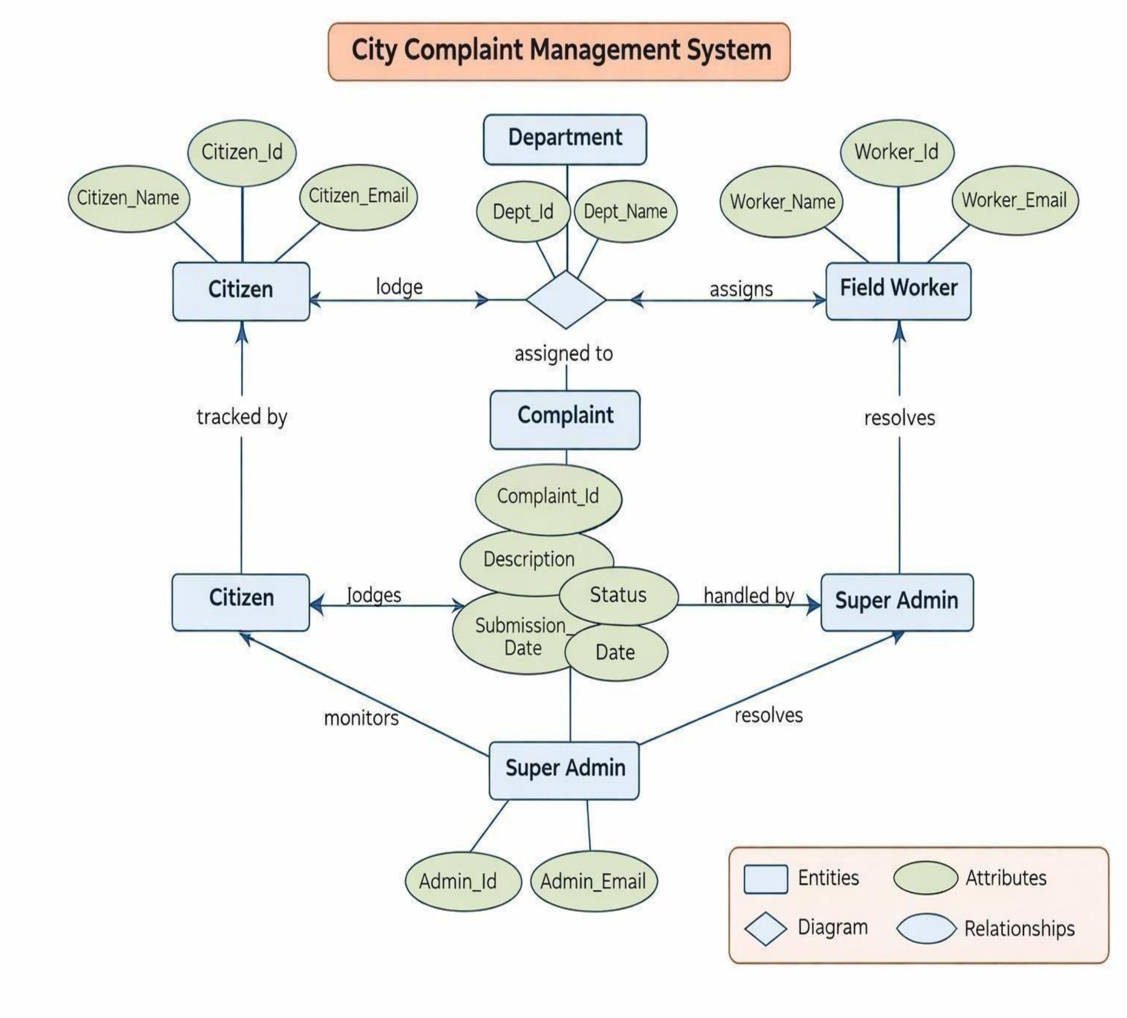

Fig: 4.1 Entity Relationship Diagram

4.2 ACTIVITY DIAGRAM

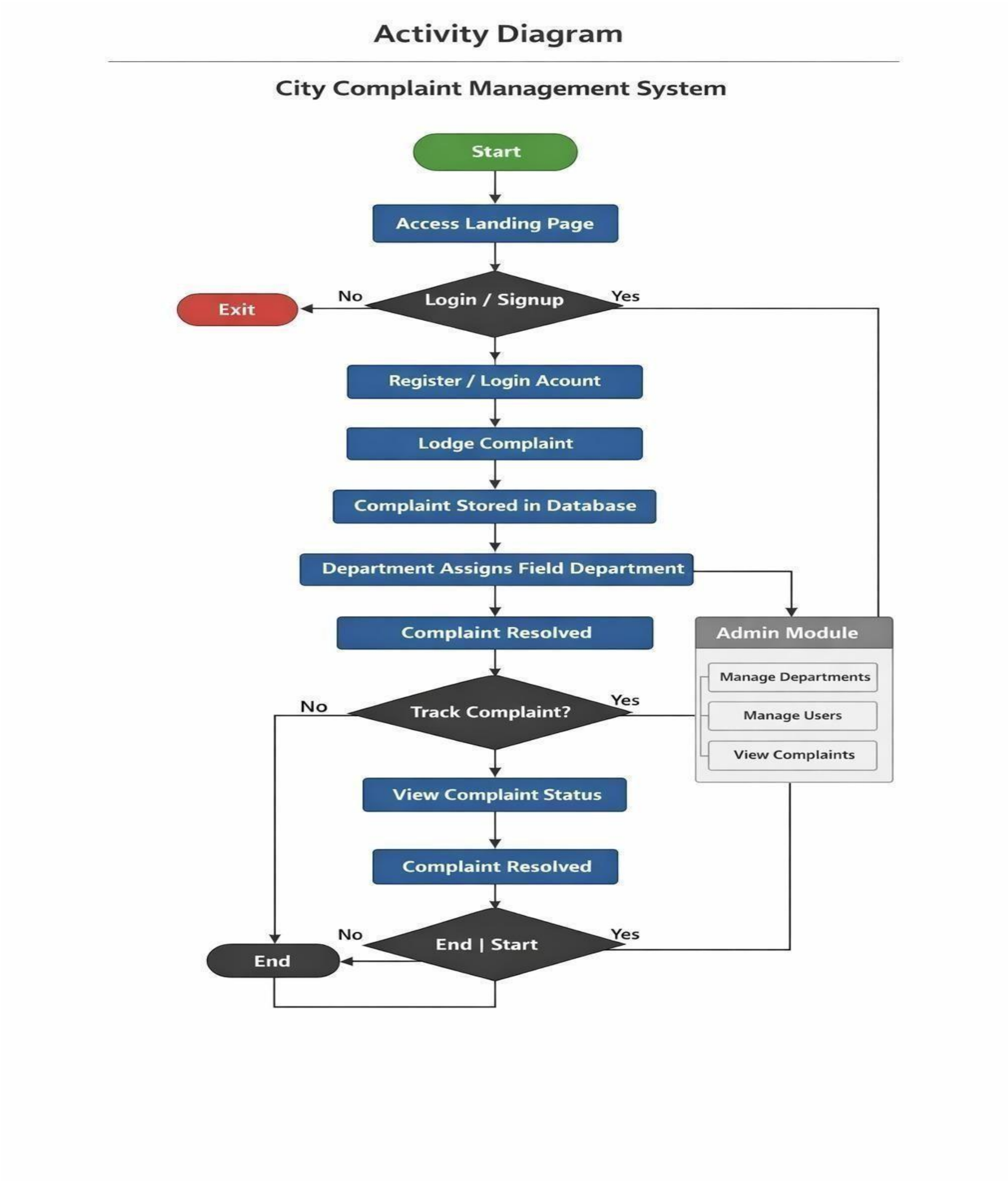

Fig: 4.2 Activity Diagram

4.3 SEQUENCE DIAGRAM

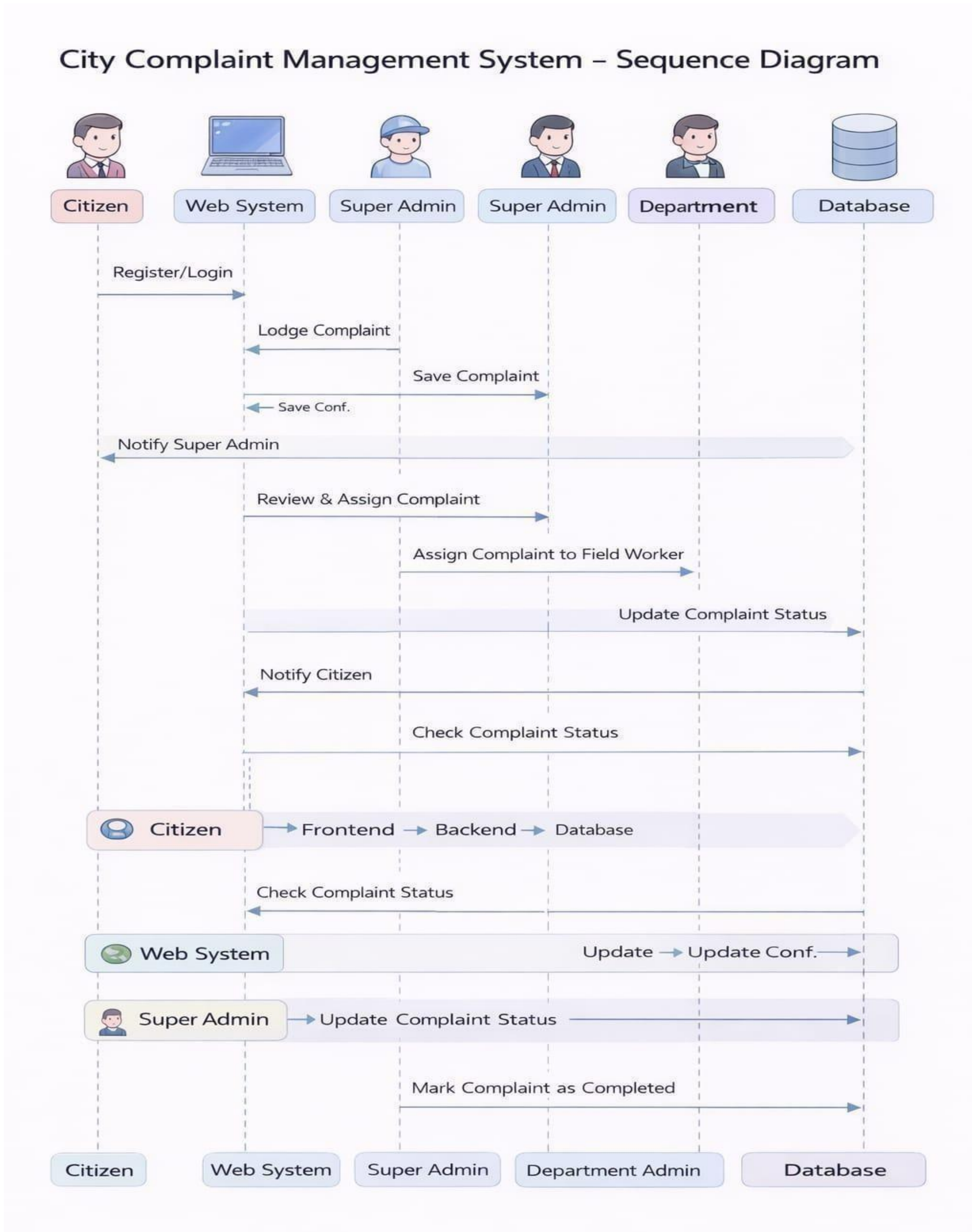

Fig: 4.3 Sequence Diagram

4.4 FLOW CHART DIAGRAM

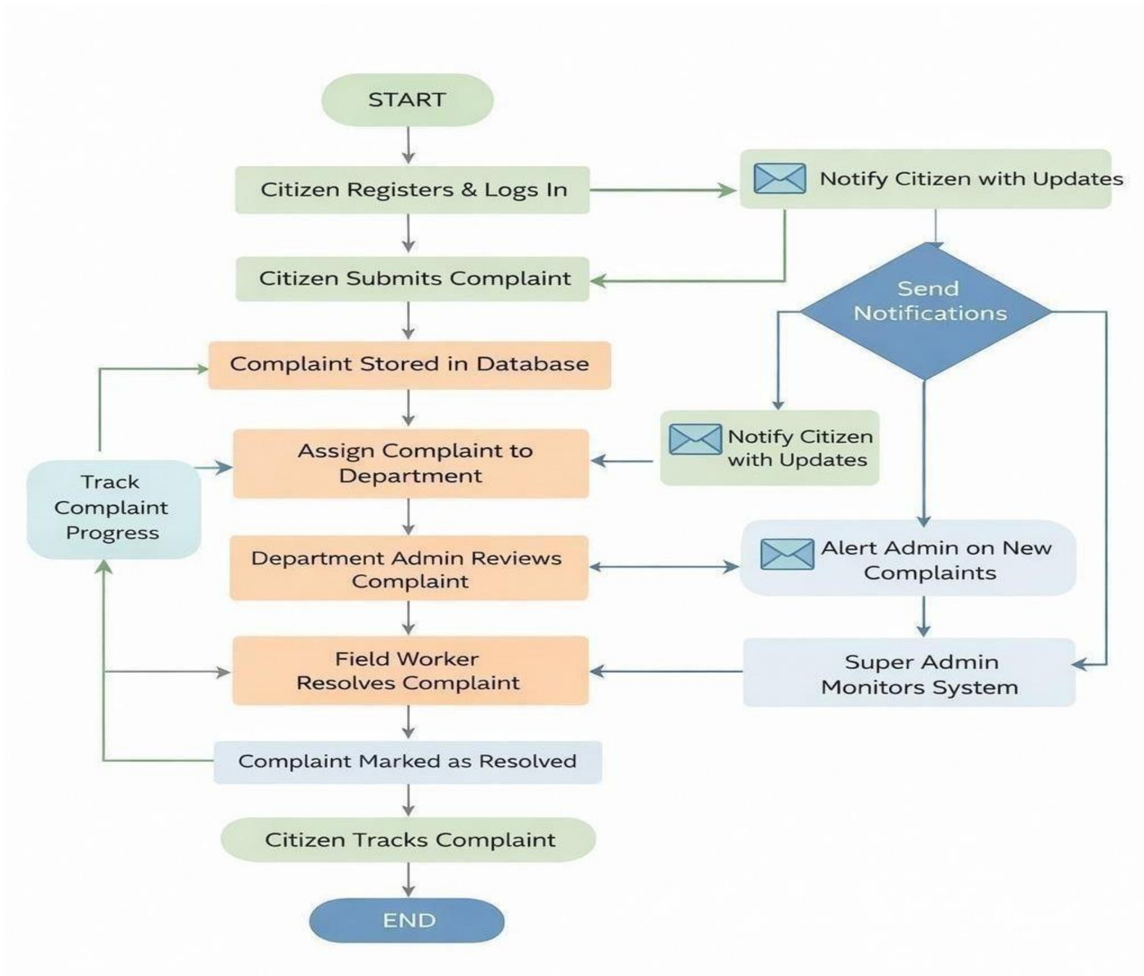

Fig: 4.4 Flow chart

4.5 SYSTEM ARCHITECTURE

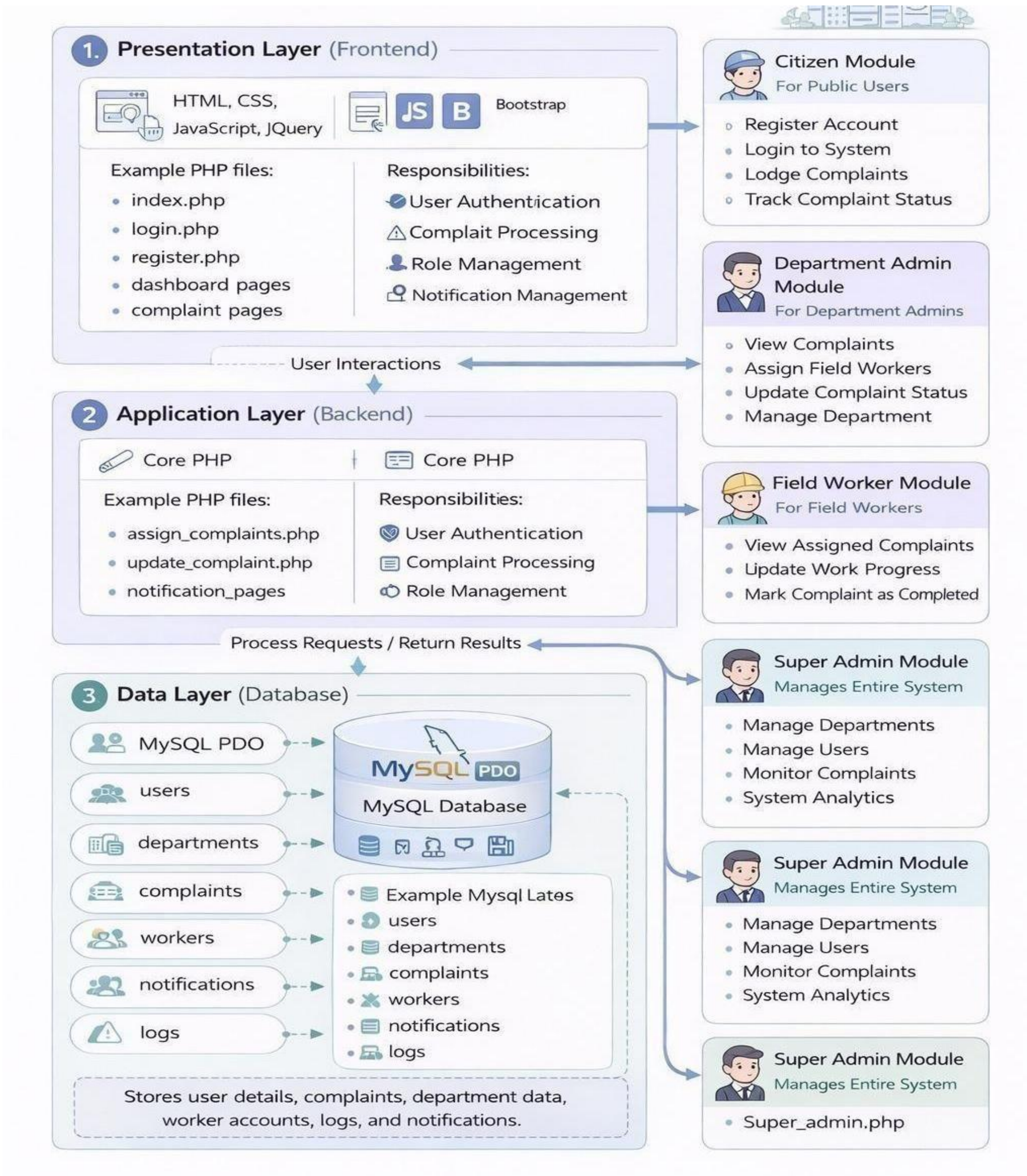

Fig: 4.5 System Architecture

4.5.1 System Architecture

System architecture refers to the overall structure and design of a system that defines how different components such as the user interface, application logic, and database interact with each other. It provides a framework for organizing and integrating all parts of the system to ensure efficient performance, scalability, and maintainability.

4.5.2 System Architecture Model

The architecture of the Smart City Complaint Management System is designed as a multi-tier architecture, which separates the system into different layers for better performance, scalability, and maintainability. This architecture ensures smooth interaction between users, application logic, and the database.

1.Presentation Layer (Frontend)

The Presentation Layer is the user interface of the system, which allows users such as citizens, department admins, field workers, and super admins to interact with the application. This layer is designed using HTML, CSS, JavaScript, and Bootstrap to provide a responsive and user-friendly interface. It enables users to perform various activities such as registering complaints, tracking complaint status, managing tasks, and viewing reports. The interface includes forms, dashboards, and navigation menus that make the system easy to use. It also ensures that users receive real-time updates and notifications regarding their complaints.

2.Application Layer (Backend / Middleware)

The Application Layer handles the core functionality and business logic of the system. It is developed using PHP and is responsible for processing user requests, managing complaint workflows, and controlling system operations. This layer manages activities such as complaint registration, assigning complaints to departments or field workers, updating complaint status, and generating reports.

3.Database Layer

The Database Layer is responsible for storing and managing all system data using MySQL. It maintains records of users, complaints, complaint categories, assignments, and status logs .This layer ensures data integrity, consistency, and secure storage. It allows efficient retrieval and updating of information, which helps in tracking complaints and generating reports. Proper relationships are maintained between tables to support smooth data flow across the system**.**

4.Integration Layer/External services

The Integration Layer of the Smart City Complaint Management System ensures seamless connection with external services to enhance the overall functionality of the system. This layer enables the system to communicate with thirdparty services for notifications, data sharing, and additional features. The system integrates with Email and SMS APIs to send real-time notifications and alerts to users regarding complaint registration, status updates, and resolution details. This helps in keeping citizens informed throughout the complaint lifecycle. Additionally, the system may use cloud storage services to store supporting files such as images, documents, or proof uploaded by users during complaint submission. This ensures efficient handling of large data and improves system performance.

The integration layer can also support location-based services (such as maps) to identify the exact location of complaints, which helps field workers reach the issue quickly. Furthermore, the system can be extended to include chatbot support for answering common user queries and providing guidance.

4.5.2 System Module

1. User Management Module

The User Management Module handles the registration, login, and management of all users in the system, including citizens, department admins, field workers, and super admins. It allows new users to create accounts by providing necessary details such as name, contact information, and password. This module also manages authentication and authorization, ensuring that users can access only the features assigned to their roles. It includes functionalities such as login validation, password recovery, profile management, and rolebased access control. This module ensures secure and organized handling of user data.

2. Complaint Registration Module

The Complaint Registration Module allows citizens to submit complaints related to various public issues such as water supply, electricity faults, sanitation, and road damage. Users can enter details like complaint category, description, location, and upload supporting images if required. Once submitted, the complaint is stored in the database and assigned a unique complaint ID for tracking purposes. This module ensures that complaints are recorded accurately and efficiently.

3. Complaint Management Module

The Complaint Management Module is mainly used by administrators to manage all registered complaints. It allows admins to view, filter, and organize complaints based on category, date, and status. Admins can review complaints, verify their validity, and take necessary actions such as assigning them to the appropriate department or field worker.

4. Complaint Assignment Module

The Complaint Assignment Module enables the admin or department authority to assign complaints to field workers or specific departments based on the nature of the issue. This module helps in distributing workload efficiently and ensures that complaints are handled by the appropriate personnel. It also keeps track of which complaint is assigned to which worker, improving accountability.

5. Status Tracking Module

The Status Tracking Module allows users to monitor the progress of their complaints in real time. Each complaint moves through different stages such as Pending, In Progress, Resolved, and Closed. Citizens can check updates using their complaint ID, while admins and field workers can update the status as work progresses. This module improves transparency and keeps users informed about the current status of their issues.

6 . Notification Module

The Notification Module is responsible for sending alerts and updates to users regarding their complaints. Notifications can be sent through email or SMS whenever there is a change in complaint status, assignment, or resolution. This module ensures timely communication between the system and users, reducing the need for manual follow-up.

7. Reporting and Analytics Module

The Reporting Module helps administrators analyze complaint data and system performance. It generates reports based on various parameters such as complaint category, number of complaints, resolution time, and department performance. These reports help authorities make better decisions, identify problem areas, and improve service quality. It also supports data visualization for better understanding of trend

CHAPTER 5

TESTING

Testing is a crucial phase in the Software Development Life Cycle (SDLC) that ensures the Smart City Complaint Management System functions correctly, efficiently, and securely. It verifies that the system meets all specified requirements and provides accurate and reliable results to users such as citizens, department admins, field workers, and super admins.

Testing ensures that the system is reliable, user-friendly, and free from major defects. It helps to confirm that all modules, including complaint registration, complaint assignment, status tracking, and notification, work together seamlessly. The main objective of testing is to identify and fix errors, bugs, or performance issues before the system is deployed for public use.

The testing process includes both functional and non-functional testing. Functional testing checks whether each feature of the system works as expected, such as submitting complaints, updating status, and generating reports. Non-functional testing focuses on system performance, security, usability, and database operations.

Since the system handles important public complaints and user data, ensuring accuracy, data security, and system reliability is very important. Proper testing improves system quality and enhances user trust.

5.1 Types of Testing

5.1.1 Unit Testing

Unit testing is performed to test individual components or modules of the system separately. Each function, such as user registration, login validation, complaint submission, and status update, is tested independently. and simplifies debugging.

5.1.2 Integration Testing

Integration testing is conducted to verify the interaction between different modules of the system. It ensures that modules such as complaint registration, assignment, and status tracking work together properly. This testing checks data flow between modules and identifies any issues that arise when components are combined.

5.1.3 System Testing

System testing evaluates the complete system as a whole to ensure that it meets all functional and non-functional requirements. It verifies that the entire application works correctly in a real-world environment. This includes testing the overall workflow of complaint handling from submission to closure.

5.1.4 Functional Testing

Functional testing ensures that all the functionalities of the Smart City Complaint Management System work according to the specified requirements. It verifies features such as user registration, login, complaint submission, complaint assignment, status tracking, and report generation. Each function is tested with valid and invalid inputs to ensure correct system behaviour. This testing confirms that the system performs all intended operations accurately and efficiently.

5.1.5 Performance Testing

Performance testing evaluates how the system behaves under different load conditions. It checks the system's response time, stability, and scalability when multiple users access the system simultaneously. This testing ensures that the system can handle a large number of complaints without slowing down or crashing, providing a smooth user experience.

5.1.6 Security Testing

Security testing is essential as the system handles sensitive user information and complaint data. It ensures that the system is protected from unauthorized access and common security threats such as SQL injection and cross-site scripting.

5.1.7 Usability Testing

Usability testing focuses on evaluating how easy and user-friendly the system is for users. It ensures that citizens, admins, and field workers can easily navigate the system, submit complaints, and track updates without confusion.This testing helps improve the overall user experience by identifying design issues and making necessary improvements.

Testing Process

The testing process of the Smart City Complaint Management System is carried out in a systematic manner to ensure that the application functions correctly and meets all user requirements. It involves multiple stages that help identify errors and improve system quality before deployment.

Requirement Analysis:

The testing process of the Smart City Complaint Management System is carried out in a systematic manner to ensure that the application functions correctly and meets all user requirements. It involves multiple stages that help identify errors and improve system quality before deployment.

Test Planning:

During test planning, a detailed test plan is prepared that defines the scope, objectives, testing methods, schedule, resources, and responsibilities. This phase ensures that testing activities are well-organized and executed efficiently.

1. **Test Case Design:**

   Specific test cases are written for each function of the system. Each test case includes input data, expected output, and actual result.

2. **Environment Setup:**

   A controlled testing environment is prepared with necessary software, databases, and configurations.

3. **Test Execution:**

   All test cases are executed. Any defect or bug identified during this phase is reported to developers for correction.

4. **Defect Tracking and Reporting:**

   All bugs are recorded in a defect tracking system with details such as severity, priority, and steps to reproduce.

5. **Retesting and Regression Testing:**

   After fixing bugs, the system is retested to ensure the errors are resolved and no new issues are introduced.

6. **Final Review and Closure:**

   Once all test cases pass successfully, a test summary report is prepared and submitted for final review.

5.2 FUNCTIONAL TESTING

Functional testing is performed to verify that all the features and functionalities of the Smart City Complaint Management System operate correctly according to the specified requirements. It focuses on validating each function of the system by providing appropriate input and checking whether the output matches the expected result. This type of testing ensures that the system behaves as intended and performs all operations accurately without any errors. In this system, functional testing is applied to major modules such as user registration, login authentication, complaint submission, complaint assignment, status tracking, notification, and report generation. For example, the registration module is tested to ensure that users can create accounts with valid details and that invalid inputs are rejected. Similarly, the login module is tested to verify proper authentication using correct credentials and denial of access for incorrect inputs.

The complaint registration module is tested to confirm that users can successfully submit complaints with all required details and that a unique complaint ID is generated. The complaint management and assignment modules are tested to ensure that administrators can view complaints and assign them to the appropriate department or field worker. The status tracking module is tested to verify that complaint status updates are correctly reflected at each stage, such as Pending, In Progress, Resolved, and Closed. Functional testing also checks whether the system correctly handles different types of inputs, including valid data, invalid data, and boundary conditions. It ensures proper data validation, error handling, and smooth interaction between different modules. This testing helps in identifying functional defects and ensures that the system is reliable, efficient, and ready for deployment.

5.2.1 Objectives of Functional Testing

To ensure that all system functionalities work correctly according to requirements and produce expected outputs for given inputs.

Requirement Analysis:

Functional requirements are studied from the SRS document to understand what needs to be tested.

1) **Test Case Design:**

Detailed test cases are prepared, specifying test inputs, expected results, and actual results.

2)Test Execution each test case manually or using automation tools to check the

application's behavior.

3) Defect Reporting:

Any deviations from expected results are reported as bugs and tracked until resolved.

4) Re-testing:

After fixing the defects, the test cases are re-executed to confirm the issues are resolved.

5) Regression Testing:

Ensures that recent code changes do not negatively impact existing functionalities..

5.3 INTERGRATION TESTING

Integration testing is performed to verify that different modules of the Smart City Complaint Management System work together correctly. It ensures that the interaction and data flow between modules such as user management, complaint registration, complaint assignment, and status tracking are functioning properly.This testing checks whether data entered in one module is correctly processed and reflected in another module. For example, when a user submits a complaint, it should be successfully stored the database and visible to the admin for assignment.

Similarly, updates made by field workers should be accurately reflected in the user's complaint status.

5.3.1 Advantages of Integration Testing

Detects interface and communication errors between modules .Ensures that the combined system performs correctly .Reduces the risk of defects in later stages of testing .Enhances reliability and stability of the final system. Validates end-to-end functionality of user workflows.

5.3.2 Challenges in Integration Testing

Difficulty in identifying which module caused an integration error. Complex dependency between modules can lead to test failures. Testing real-time data flow between multiple systems like payment gateways or APIs .Requires careful planning and detailed interface documentation.

5.3.3 Tools Used for Integration Testing

- **Postman:** To test APIs and data exchange between modules.
- **`Selenium:** To automate integration test scenarios.
- **JUnit / PyTest:** For backend integration logic testing.
- **JMeter:** To test integrated system performance under load.

5.4 PERFORMANCE TESTING

Performance testing is conducted to evaluate how efficiently the Smart City Complaint Management System operates under different conditions. It checks the system's response time, speed, stability, and ability to handle multiple users simultaneously.This testing ensures that the system can process a large number of complaints without delays or failures. It also verifies that pages load quickly, database operations are efficient, and the system remains stable even during peak usage. Performance testing helps in identifying bottlenecks and improving overall system efficiency.

5.5 USABILITY TESTING

Usability testing is conducted to evaluate how easy and user-friendly the Smart City Complaint Management System is for different types of users such as citizens, department admins, field workers, and super admins. The main purpose of this testing is to ensure that users can interact with the system effectively without confusion or difficulty. It focuses on the overall user experience, including navigation, interface design, clarity of instructions, and ease of performing tasks.

In this system, usability testing is applied to key features such as user registration, login, complaint submission, complaint tracking, and status updates. It ensures that forms are simple to understand, buttons and menus are clearly labeled, and users can complete tasks with minimal effort. The system interface is designed using Bootstrap to provide a clean, responsive, and consistent layout across different devices such as desktops and mobile phones.

Usability testing also checks whether error messages are meaningful and guide users to correct mistakes easily. For example, if a user enters invalid data during registration or complaint submission, the system should display clear instructions to fix the issue. It also ensures that users can easily track their complaints and understand the status updates without confusion. This testing involves real users interacting with the system and providing feedback on their experience. Based on this feedback, necessary improvements are made to enhance usability, accessibility, and overall user satisfaction. By ensuring a smooth and intuitive interface, usability testing plays a key role in making the system efficient and widely acceptable.

5.6 SECURITY TESTING

Security testing is performed to ensure that the Smart City Complaint Management System is protected against unauthorized access, data breaches, and various security threats. Since the system handles sensitive user information and complaint data, maintaining data confidentiality, integrity, and availability is very important.

This testing focuses on verifying the security mechanisms implemented in the system, such as user authentication, authorization, and data protection. It ensures that only valid users can access the system using proper login credentials. Role-based access control is also tested to confirm that each user (citizen, admin, field worker, super admin) can access only the features permitted to them.

CHAPTER 6

SYSTEM IMPLEMENTATION

System implementation is the phase where the Smart City Complaint Management System is developed and deployed for real-time use. The system is built using PHP, MySQL, HTML, CSS, JavaScript, and Bootstrap. The database is created to store user details, complaints, and status updates.

The front-end provides a user-friendly interface for citizens and admins. The backend handles data processing, validation, and communication with the database. All modules such as registration, login, complaint submission, assignment, and tracking are implemented. The system is tested to ensure proper functionality and performance. Security measures like input validation and password protection are applied. Finally, the system is deployed and made accessible through a web browser for users.

6.1 Objectives of Implementation

The main objective of system implementation is to convert the system design into a fully functional application. It ensures that all modules are properly developed, integrated, and deployed for real-time use. The goal is to provide a reliable, efficient, and user-friendly system that meets all user requirements.

1.Successful System Deployment

Ensures the system is properly installed, configured, and tested before being made available to users. It confirms that all modules are working correctly in the live environment without major errors.

2. User Adoption and Proficiency

Ensures that users can easily understand and effectively use the system. Proper guidance and simple interface help users perform tasks like complaint submission and tracking without difficulty.

3. Data Integrity and Security

Ensures that all data stored in the system is accurate, consistent, and protected. Security measures like validation, authentication, and access control prevent unauthorized access and data loss.

6.2 SAMPLE SCREEN CAPTURES

6.2.1 Log In /Register Page

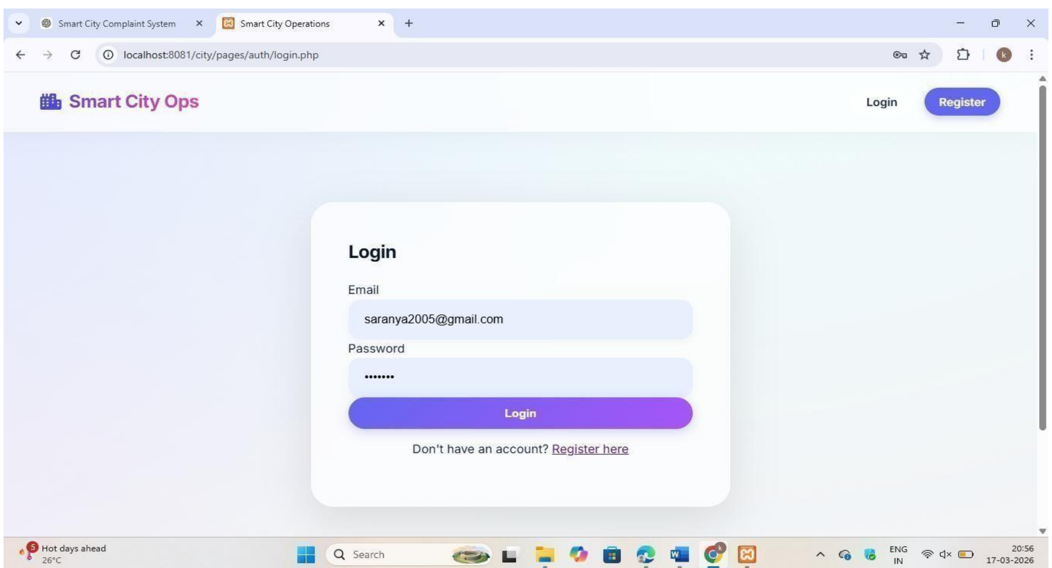

Fig: 6.2.1 Log In /Register Page

6.2.2 Citizen Page

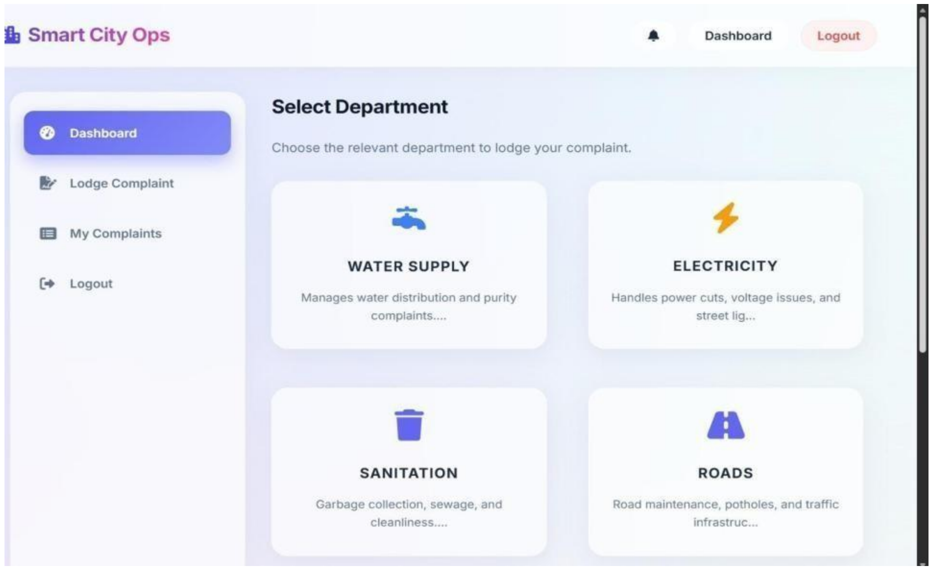

Fig: 6.2.2 Citizen Page

6.2.3 Water Admin Page

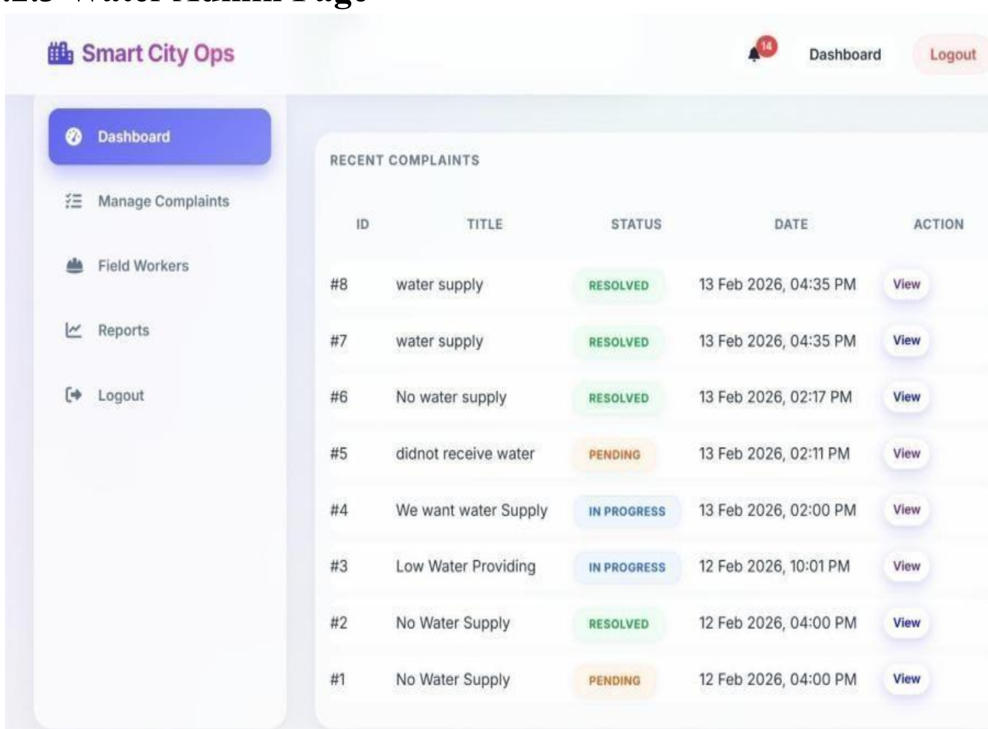

Fig: 6.2.3 Water Admin page

6.2.4 Field Worker

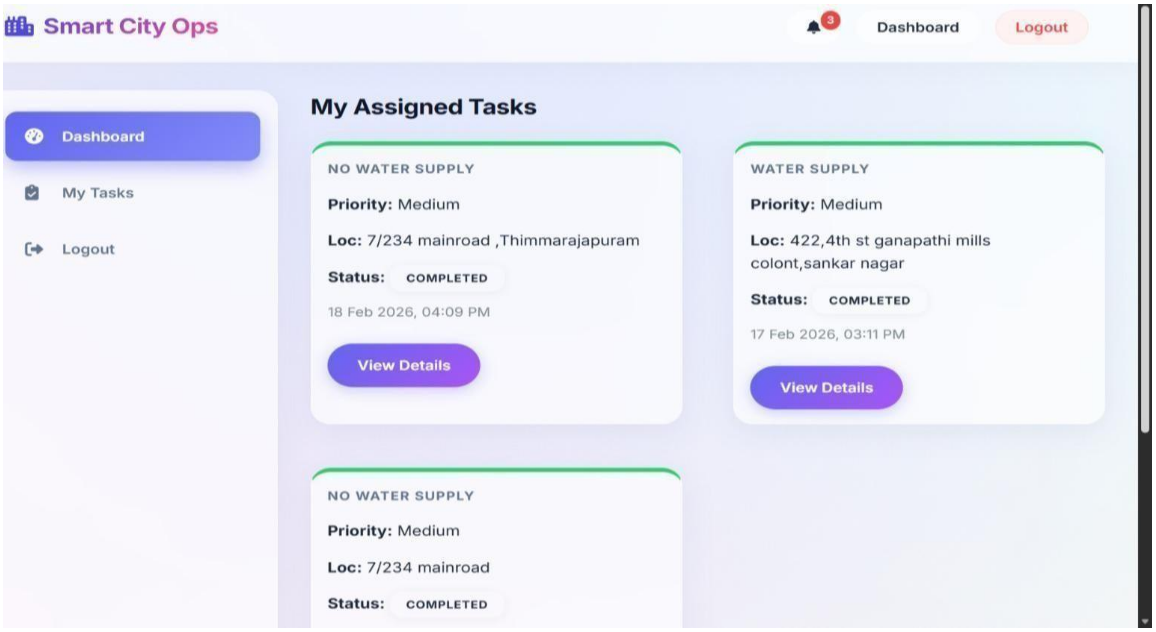

Fig: 6.2.4 Field Worker

6.2.5 MAIN ADMIN REPORT PAGE

Fig: 6.2.5 Main Admin Report page

6.3 SAMPLE SOURCE CODE

6.3.1 LOG IN .PHP

```
<?php
// pages/auth/login.php require
'../../config/db.php'; require
'../../includes/functions.php';
$error = '';
if ($_SERVER["REQUEST_METHOD"] == "POST") {
$email = sanitize($_POST['email']);
$password = trim($_POST['password']); // Trim whitespace!
if (empty($email) || empty($password)) {
  $error = "Please fill all fields.";
} else {
$stmt = $pdo->prepare("SELECT u.*, r.name as role_name FROM users u JOIN
roles r ON u.role_id = r.id WHERE u.email = ?");
$stmt->execute([$email]);
$user = $stmt->fetch();
if ($user && password_verify($password, $user['password'])) {
session_regenerate_id(true); // Prevent Session Fixation
$_SESSION['user_id'] = $user['id'];
$_SESSION['role_id'] = $user['role_id'];
$_SESSION['role_name'] = $user['role_name']; // 'Super Admin', 'Citizen', etc.
$_SESSION['dept_id'] = $user['dept_id'];
```

```
$_SESSION['full_name'] = $user['full_name'];
// Redirect based on role            $role_name =
$user['role_name'];            if ($role_name == 'Super
Admin') {                header("Location:
../dashboard/super_admin.php");           } elseif
($role_name == 'Department Admin') {
header("Location: ../dashboard/dept_admin.php"); }
elseif ($role_name == 'Field Worker') {
header("Location: ../dashboard/field_worker.php");
} else {                header("Location:
../dashboard/citizen.php");
}
 exit();
} else
{

$error = "Invalid email or password.";
}
}
}
?>

<?php include '../../includes/header.php'; ?>
<div class="auth-container">
<h2>Login</h2>
```

```
<?php if($error): ?>
<div class="alert alert-error"><?php echo $error; ?></div>
<?php endif; ?>
<form method="POST" action="" autocomplete="off">
<div class="form-group">
<label class="form-label">Email</label>
<input type="email" name="email" class="form-control" required value="">
</div>
<div class="form-group">
<label class="form-label">Password</label>
<input type="password" name="password" class="form-control" required>
</div>
<button type="submit" class="btn btn-primary">Login</button>
</form>
<p style="margin-top: 15px; text-align: center;">
Don't have an account? <a href="register.php">Register here</a>
  </p>
</div>
<?php include '../../includes/footer.php'; ?>
```

6.3.2 LOG IN.HTML

```
<!DOCTYPE html>
<html>
<head>
```

```
<title>City Complaint Login</title>
<style>        body{            font-
family: Arial, sans-
serif;
background:#f4f6f9;
}
.login-box{            width:350px;
margin:80px auto;
background:#fff;            padding:30px;
borderradius:10px;            box-
shadow:0 0
10px rgba(0,0,0,0.1);
} h2{
textalign:center;
marginbottom:2
0px; }
input{
width:100%;
padding:10px;
margin:10px
0;
```

```
}       button{
width:100%;
padding:10px;
background:#007bff
;
border:none;
color:#fff;
cursor:pointer; }
button:hover{
background:#0056b3;
}       .error{
color:red;
textalign:center;
}
</style>
</head>
<body
<div class="login-box">
<h2>Login</h2>
<?php if(!empty($error)) { ?>
<p class="error"><?php echo $error; ?></p>
```

```
<?php } ?>

<form method="POST" action="">

<input type="email" name="email" placeholder="Enter Email" required>

<input type="password" name="password" placeholder="Enter Password" required>

<button type="submit" name="login">Login</button>

</form>

</div>

</body>

</html>
```

6.3.3 DATABASE CODE

```
CREATE TABLE users (     id INT

AUTO_INCREMENT PRIMARY KEY,     name

VARCHAR(100) NOT NULL,     email

VARCHAR(100) NOT NULL UNIQUE,     password

VARCHAR(255) NOT NULL,     role

VARCHAR(20) DEFAULT 'active',     created_at

TIMESTAMP DEFAULT

CURRENT_TIMESTAMP

);
```

CHAPTER 7

CONCLUSION

The Smart City Complaint Management System successfully provides an efficient and reliable solution for handling public grievances in urban areas. It simplifies the process of complaint registration, tracking, and resolution, making it easier for both citizens and authorities to manage issues effectively. By replacing traditional manual methods, the system ensures faster response times and improved service delivery.The system enhances communication through a transparent and user-friendly platform, where features like realtime tracking, notifications, and role-based access increase accountability and user satisfaction. It also improves data management, reduces errors, and saves time and resources by utilizing modern web technologies.

Furthermore, the system supports better decision-making through reports and analytics, helping authorities identify recurring problems and take preventive measures. With future enhancements such as mobile applications and AI-based analysis, the system can become even more powerful and scalable.

In addition, the system encourages active citizen participation by providing an easy and accessible platform for raising concerns. It strengthens the relationship between the public and government authorities by ensuring that every complaint is acknowledged and addressed properly. This not only improves service quality but also promotes transparency, responsibility, and trust in governance. Overall, the Smart City Complaint Management System contributes to digital governance and plays a key role in building a smarter, more efficient, and citizen-friendly city environment.

CHAPTER 8

FUTURE ENHANCEMENT

The Smart City Complaint Management System can be further improved by incorporating advanced features and modern technologies to enhance its functionality, usability, and overall efficiency. One of the major enhancements is the development of a dedicated mobile application for Android and iOS platforms, which will enable users to register and track complaints conveniently from their smartphones anytime and anywhere.

The system can also be integrated with GPS and location-based services to automatically detect and tag the exact location of the complaint. This will help field workers quickly identify the problem area and provide faster and more accurate resolutions. Additionally, incorporating AI-based analysis and image processing can help in automatically categorizing complaints based on uploaded images, reducing manual effort and improving accuracy.Another important enhancement is the integration of realtime chat or chatbot support, which can assist users by answering their queries and guiding them through the complaint registration process. This will improve user interaction and make the system more user-friendly, especially for new users.

CHAPTER 9
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